(Please write your Enrollment Number) Enrollment No. 02701182025

End-Term Examination
(CBCS);SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech>, Semester:<1st>
(December, 2025)

Subject Code: BEC 103	Subject: Electronics Workshop
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8 =20)	CO Mapping
	a) What is the operating point (Q-point) of a transistor? What is its significance?		
	b) Define cut-off and saturation regions in a BJT.		
	c) Define stabilization factor and explain its significance in bias stabilization.		
	d) Write short note on Electrodynamometer Type Wattmeter.		
	e) Define Vector Voltmeter and state its main applications.		
	f) What determines the resolution and accuracy of frequency measurement?		
	g) Convert the following expression in canonical POS form. $Y = (A + B)(\bar{B} + C)$		
	h) Simplify using Boolean Algebra $Y = AB + \overline{AC} + A\bar{B}C\,(AB + C)$		
	UNIT I		**CO Mapping**
Q2	A silicon diode has a reverse saturation current (IS) of 10^{-12} A Find the forward current (IF) when the forward voltage V_F=0.7 V at 27°C. Use the diode equation: $I = I_S\left(e^{\frac{V}{\eta V_T}} - 1\right)$ where V_T=26 mV and η=1.	**(10)**	
Q3	(a) Explain the construction, operation and characteristics of a BJT in CE configuration. (b) Derive the relationship between α and β, and mention typical values of these parameters.	**(10)**	
	UNIT II		**CO Mapping**
Q4	A transistor circuit uses a voltage-divider bias with the following parameters: V_{CC}=12V, R_1=100kΩ, R_2=20kΩ, R_C=2kΩ, R_E=1kΩ, β=100	(10)	

	Calculate: (a) Base voltage V_B (b) Emitter voltage V_E (c) Collector current I_C (d) Collector-emitter voltage V_{CE}		
Q5	Explain the need for bias stabilization in transistor amplifiers. Discuss the biasing scheme: voltage-divider bias with neat circuit diagram. (b) Derive expression for stabilization factor with respect to I_{CO}, VBE, and β.	(10)	
	UNIT III		**CO Mapping**
Q6	Explain the construction and working principle of a Digital Multimeter (DMM). Discuss how it measures voltage and current.	(10)	
Q7	A True RMS Voltmeter gives a reading of 10 V RMS when connected to a waveform consisting of a 5 V DC component and an AC component. Calculate: (a) The RMS value of the AC component, and (b) The average value of the waveform (assume the AC component is sinusoidal).	(10)	
	UNIT IV		**CO Mapping**
Q8	Use K-Maps to minimize and express the function in SOP form. Implement the expression using NAND gates only. $F = \sum m (0, 1, 2, 6, 7, 10, 12, 15) + \sum d (3, 8, 13, 14)$	(10)	
Q9	Implement the following three Boolean functions using a 3:8 decoder constructed with NAND gates (active low o/ps). Use block diagram for the decoder. F1 (A B C) =$\sum$ m (2, 4, 7) F2(A B C) =$\sum$ m (0, 3) F3 (A B C) =$\sum$ m (0, 2, 3, 4, 7)	(10)	